Rhutam Mahajan

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Education

University of Maryland, College Park — Aug 2025 – May 2027
M.S. in Data Science | GPA: 3.95/4.0 — *College Park, MD*

Savitribai Phule Pune University — Dec 2021 – Jun 2025
B.E. in Computer Engineering, Honors in Data Science | GPA: 8.5/10 — *Pune, India*

Technical Skills

Languages & Core Libraries: Python, SQL, Pandas, NumPy, SciPy, scikit-learn
Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, Hugging Face Transformers, computer vision, NLP, few shot and multimodal learning, time series forecasting
Generative AI & LLMs: OpenAI Agents SDK, Model Context Protocol (FastMCP), LangChain, LangGraph, ChromaDB, RAG, tool calling
Big Data, Cloud & Tools: Apache Spark, PySpark, Spark MLlib, Neo4j, MySQL, AWS (ECR, S3, App Runner), Docker, FastAPI, Git, Tableau, Power BI

Experience

InternPe — Feb 2024 – Mar 2024
AI/ML Intern — *Remote*

- Built four end to end machine learning pipelines covering binary, multiclass, and regression tasks with Python, scikit-learn, TensorFlow, and Pandas.
- Raised match prediction accuracy from 71.11% to 94.87% by benchmarking Logistic Regression against Random Forest and engineering live game state features across 149,578 IPL records.
- Trained models on datasets up to 179,078 records, reaching 96.49% accuracy on a breast cancer neural network and an R^2 of 0.9210 on car price regression.

Gamaka AI — Dec 2023 – Mar 2024
Junior Data Scientist — *Pune, India*

- Designed normalized MySQL schemas with up to 12 related tables, enforcing primary and foreign key integrity, and profiled source data to resolve missing values, duplicates, and outliers before analysis.
- Wrote analytical SQL spanning multi table joins, grouping, and year over year comparisons to report revenue, expense, profit, and earnings per share trends across seven companies of financial records.
- Developed and validated classification, regression, clustering, and time series models in Python with scikit-learn, TensorFlow, and Keras, reaching 99% accuracy on SMS spam detection and 97% on handwritten digit recognition.
- Scored sentiment on over 20,000 hotel reviews and shipped interactive Tableau and Power BI dashboards on airport traffic, retail sales, and banking activity to surface trends for non technical stakeholders.

Projects

Agentic AI Systems with MCP, RAG, and Evaluator Loops — Apr 2026
OpenAI Agents SDK, MCP (FastMCP), LangGraph, LangChain, ChromaDB

- Built async multi-agent workflows in Python spanning research, tool calling, memory, retrieval, and evaluation over MCP servers.
- Wrote a custom FastMCP server exposing tools to read state, execute actions, update strategy, and serve resources to agents.
- Built a RAG chatbot with document ingestion and Chroma vector search, using OpenAI for generation and Gemini for evaluation.

Spark Readmission Pipeline & Graph Data Engineering — Apr 2026
PySpark, Spark MLlib, Neo4j

- Built a PySpark cleaning and modeling pipeline on 12,465 readmission records, handling missing values, mismatches, and outliers.
- Engineered eight numeric and eight categorical features through Spark ML pipelines (StringIndexer, OneHotEncoder, VectorAssembler) and trained a Spark MLlib Random Forest (100 trees, max depth 8) on an 80/20 split.
- Scored 60.57% accuracy, 58.29% F1, 61.60% precision, and 0.636 AUC on held out data.

Multimodal Medical Triage — Dec 2025 – Mar 2026
PyTorch, FastAPI, Docker, GitHub Actions, AWS, React TypeScript

- Built a multimodal triage model that fuses lesion images (ResNet-18 encoder) with clinical text (128 dim embedding encoder) through a shared classifier head to predict Low, Medium, or High risk.
- Trained on 5,067 cases with an 80/20 stratified split using class weighted cross entropy and AdamW, reaching 80.13% accuracy and 77.32% macro F1.
- Built a FastAPI inference service with a React and TypeScript frontend, Dockerized the backend, and deployed to AWS (ECR, App Runner, S3) with GitHub Actions CI/CD.

Plant Leaf Disease Classification via Few-Shot Learning — Aug 2024 – May 2025
EfficientNet-B0, Prototypical Networks, PyTorch

- Built a two level few-shot classifier on EfficientNet-B0, collapsing 38 raw classes into 12 disease categories.
- Trained a 384 dim EfficientProtoNet on balanced few-shot splits (240 images) and evaluated on 13,135 held out PlantVillage images: 97.77% accuracy, 95.46% macro precision, and 95.83% macro F1.

Publications

Plant Leaf Disease Multilevel Classification Using Few-Shot Learning — May 2025
International Journal of Research and Analytical Reviews